

Edward Pearson Chandler

epchandler@wisc.edu · edchandler00.github.io · LinkedIn · Google Scholar

Research Area

Computational Imaging, Deep Learning, Optimization, Inverse Problems, and Computer Vision

Education

University of Wisconsin–Madison <i>Ph.D. in Electrical and Computer Engineering</i> • Advisor: Prof. Ulugbek S. Kamilov	2026 – Present GPA: 4.00 / 4.00
Washington University in St. Louis <i>Ph.D. Student in Computer Science</i> (Transferred with Advisor) • Advisor: Prof. Ulugbek S. Kamilov	2023 – 2025 GPA: 4.00 / 4.00
Washington University in St. Louis <i>B.S. in Computer Science + Mathematics</i> • Minor: Philosophy-Neuroscience-Psychology • Research Advisor: Prof. Ulugbek S. Kamilov • Academic Advisor: Prof. Ron Cytron	2019 – 2023 GPA: 4.00 / 4.00

Awards

NSF GRFP Honorable Mention (2024)
McKelvey School of Engineering Valedictorian (2023)
McKelvey School of Engineering Award for Research Excellence – CSE (2023)
McKelvey Dean’s Select PhD Fellowship (2022)
Antoinette Frances Dames Award for Productive Scholarship in Engineering (2021)

Technical & Relevant Experience

Computational Imaging Group – <i>Graduate Researcher</i> • Develop deep learning and convex/non-convex optimization algorithms to solve high-dimensional inverse problems.	Sep. 2020 – Present
Los Alamos National Laboratory – <i>Graduate Researcher</i> • Designed and implemented large-scale Computed Tomography/Laminography image reconstruction algorithms.	Jun. 2025 – Aug. 2025, May 2026 – Present
ENS Paris (Complex Media Optics Lab) – <i>Visiting Graduate Researcher</i> • Collaborated on Raman imaging system modeling, leading to a publication in <i>Nature Communications</i> .	Jan. 2024 – May 2024
General Dynamics Information Technology – <i>Computer Vision Data Science Intern</i> • Developed deep learning segmentation models for satellite imagery datasets.	Sep. 2021 – Aug. 2022
Washington University in St. Louis – <i>Teaching Assistant (DSA)</i> • Led weekly group tutorials on analytical and coding problems.	Sep. 2020 – Dec. 2020
The Trade Desk – <i>Software Engineer Intern</i> • Built a full-stack news app using The Trade Desk’s database of internet users.	Jan. 2019 – Apr. 2019

Scholarly Activities

Conferences Reviewed

- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- European Conference on Computer Vision (ECCV) 2026
- Conference on Neural Information Processing Systems (NeurIPS) 2026

Journals Reviewed

- IEEE Transactions on Computational Imaging (TCI)
- IEEE Signal Processing Letters (SPL)
- SIAM Journal on Imaging Sciences (SIIMS)

Publications (* Equal contribution/Shared first authorship)

- [1] S. Shoushtari, **E. P. Chandler**, and U. S. Kamilov, “Moments Matter: Posterior Recovery in Poisson Denoising via Log-Networks,” *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2026.
- [2] A. Scott, E. Louwagie, **E. P. Chandler**, S. Kolluru, C. Foshier, A. Arter, F. Manyx, U. S. Kamilov, A. Odibo, and M. Oyen, “Predicting Stillbirth Risk Factors with Ultrasound Texture Analysis and Doppler Imaging of the Placenta,” *Reproductive Sciences*, 2026.
- [3] **E. P. Chandler**, S. Shoushtari, B. Wohlberg, and U. S. Kamilov, “Closed-Form Approximation of the Total Variation Proximal Operator,” *IEEE Transactions on Computational Imaging (TCI)*, 2025.
- [4] S. Zhao, L. Chibani, **E. P. Chandler**, F. Liu, J. Hu, L. Valzania, U. S. Kamilov, and H. B. de Aguiar, “Computational field-resolved coherent chemical imaging,” *Nature Communications*, 2025.
- [5] S. Shoushtari, J. Liu, **E. P. Chandler**, M. S. Asif, and U. S. Kamilov, “Prior Mismatch and Adaptation in PnP-ADMM with a Nonconvex Convergence Analysis,” *Proc. International Conference on Machine Learning (ICML)*, 2024.
- [6] **E. P. Chandler**, S. Shoushtari, J. Liu, M. S. Asif, and U. S. Kamilov, “Overcoming Distribution Shifts in Plug-and-Play Methods with Test-Time Training,” *IEEE Int. Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, 2023.
- [7] W. Wang*, S. Shoushtari*, **E. P. Chandler**, M. S. Asif, and U. S. Kamilov, “Distribution Shift Estimation in Imaging Inverse Problems,” *Proc. IEEE Int. Workshop Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, 2025.

Preprints (* Equal contribution/Shared first authorship)

- [1] X. Shi*, **E. P. Chandler***, C. Y. Park, S. Shoushtari, and U. S. Kamilov, “NullFlow: One-Step Generative Reconstruction,” *arXiv preprint arXiv:2606.22696*, 2026. (*Co-first authors)
- [2] C. Y. Park, **E. P. Chandler**, Y. Hu, M. T. McCann, C. Garcia-Cardona, B. Wohlberg, and U. S. Kamilov, “Stochastic Generative Plug-and-Play Priors,” *arXiv preprint arXiv:2604.03603*, 2026.
- [3] S. Shoushtari, **E. P. Chandler**, Y. Wang, M. S. Asif, and U. S. Kamilov, “Unsupervised Detection of Distribution Shift in Inverse Problems using Diffusion Models,” *arXiv preprint arXiv:2505.11482*, 2025.
- [4] **E. P. Chandler**, S. Shoushtari, B. Wohlberg, and U. S. Kamilov, “Analysis Plug-and-Play Methods for Imaging Inverse Problems,” *arXiv preprint arXiv:2509.15422*, 2025.

Technical Skills

Programming & Tools: Python (PyTorch, NumPy), MATLAB, Linux/Bash, Git, Docker

Domain Knowledge: Computational Imaging, Deep Learning, Optimization, Inverse Problems, and Computer Vision